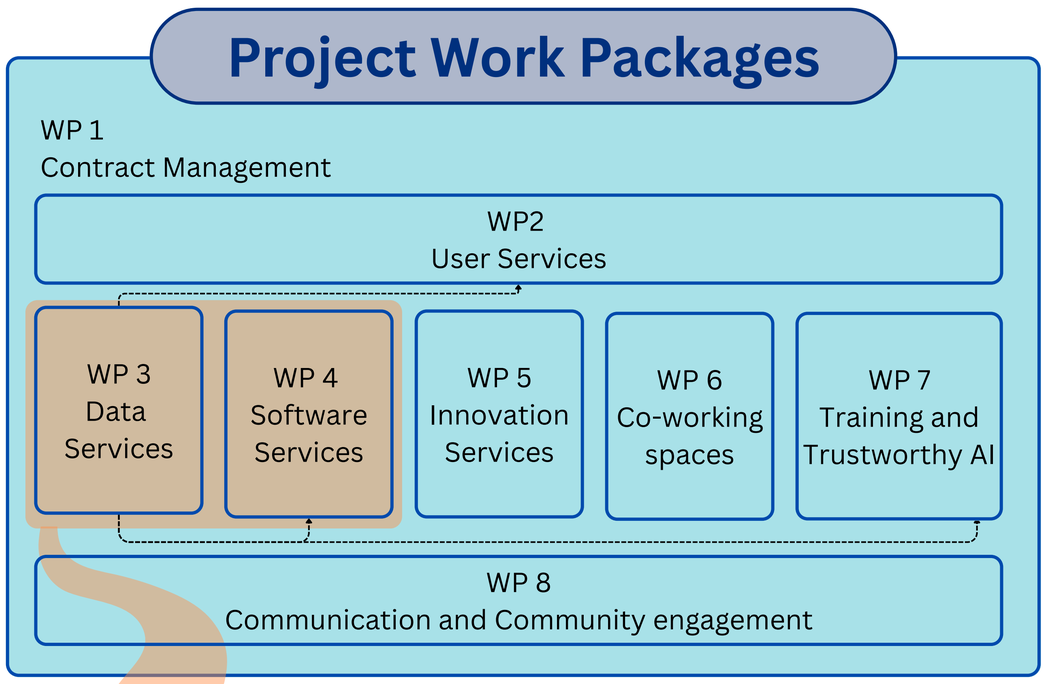
Empowering Responsible AI: Data & Software Management at BSC AI Factory

Aina Jené Cortada[1], Eudald Lerga[1], Nadia Tonello[1], Mercè Crosas[1]
[1]*Barcelona Supercomputing Center (BSC)*

What's the BSC AI Factory?

The BSC AI Factory project offers a comprehensive suite of Artificial Intelligence (AI) services, training, and networking hubs designed to accelerate AI adoption by European industries—especially small-medium enterprises (SME) and startups—across four countries. It is powered by one of Europe's most powerful computing systems, MareNostrum5, a supercomputer built to handle massive AI workloads and advanced data processing, helping meet Europe's growing needs for AI innovation.

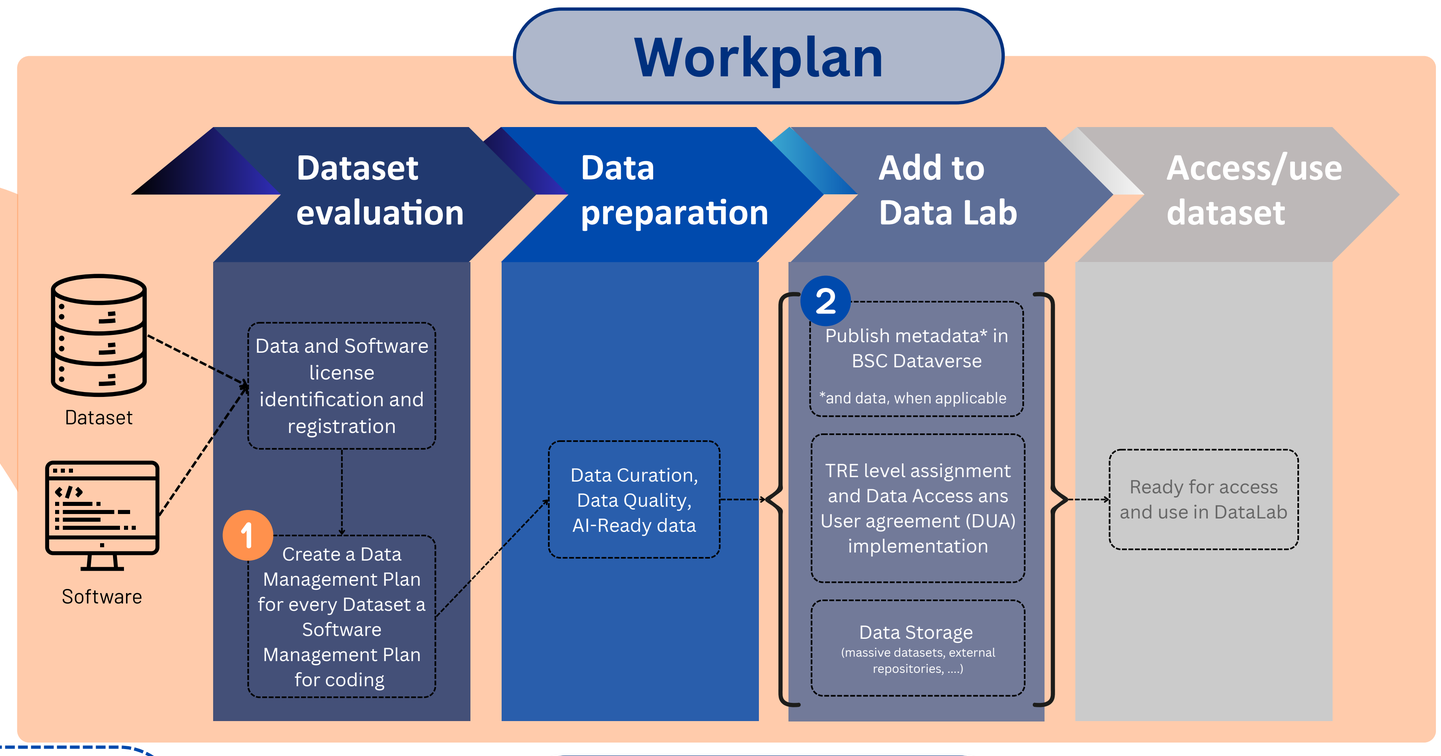
Project Work Packages



Core Objectives

- Accelerate AI use in SMEs, startups, academia, and public institutions.
- Promote FAIR, open, and sustainable data and software practices.
- Train diverse users to close the AI skills gap.
- Foster ethical, inclusive, and impactful AI innovation.
- Build a cross-border AI ecosystem with shared hubs and tools.

Workplan



1 Data Stewardship Wizard

DSW
DATA STEWARDSHIP WIZARD

Why Data Management Plans?
Ensure data is FAIR, open, and responsibly managed.

Our Approach:
Require a Data Management Plan (DMP) for every dataset. And Software Management Plan (SMP) for coding.

Why use Data Stewardship Wizard (DSW)?

- Standardised, easy-to-follow questionnaire
- Version control for ongoing updates
- Question tagging and filtering for relevance
- Built-in metrics to assess FAIRness and openness

Benefits:

- Clear, transparent data and software management
- Compliance with open science and ethical standards
- Allows collaboration

GitLab

2 Dataverse

Why using a repository?
Organise and provide easy access to all datasets and related software within the AI Factory.

Our Approach:
Register every dataset and software asset in a unified repository system combining BSC Dataverse and GitLab.

Why BSC Dataverse?

- Centralised, searchable data catalogue
- Integration with GitLab for software versioning and management
- Links with Data Stewardship Wizard to check DMPs and SMPs
- Supports secure, ethical, and compliant data/software sharing

Benefits:

- Streamlined access to data and code for researchers and industry
- Enhanced reproducibility and transparency
- Facilitates collaboration across sectors and countries
- Aligns with ethical and legal frameworks for AI innovation

The Dataverse Project

Conclusion

- The BSC AI Factory ensures responsible, FAIR, and open data and software management.
- With local instances of DSW, Dataverse, and GitLab, we enable ethical, transparent, and reproducible AI.

GENERACIÓN D · Barcelona Supercomputing Center · AI4S BSC AI4Science Fellowships · elixir SPAIN · Financiado por la Unión Europea NextGenerationEU · red.es · Plan de Recuperación, Transformación y Resiliencia · EuroHPC Joint Undertaking